

7.3 Trigonometric Substitution

How do we relate the sides of a right triangle?

Suppose we wish to evaluate

$$\int \sqrt{9 + x^2} \, dx$$

Will previously learned techniques suffice?

Use properties of triangles to evaluate

$$\int \sqrt{9 + x^2} \, dx$$

Give the range of the following inverse trigonometric functions.

1. $\arcsin(x) = \sin^{-1}(x)$

2. $\arctan(x) = \tan^{-1}(x)$

3. $\operatorname{arcsec}(x) = \sec^{-1}(x)$

Procedure for Trigonometric Substitution

1. Draw a right triangle with appropriate side lengths and angle θ labeled. Write down $\sin(\theta)$, $\cos(\theta)$, and $\tan(\theta)$, in terms of the side lengths.
2. Write down the substitution for the variable _____, calculate the differential _____, and specify the selected values of _____ for substitution.
3. _____ the trigonometric expression and the calculated _____ into the integrand. Simplify the results algebraically.
4. _____ the trigonometric integral, keeping in mind the restrictions on the angle θ for reversibility.
5. Use the _____ to reverse the substitution in the integration result and convert it back to the original variable x . In particular, compositions of trig and inverse trig functions of x should be avoided (final answers should be in fully simplified form).

Example 1

Evaluate $\int \frac{x^3}{\sqrt{16+x^2}} dx$

Extra Example

Evaluate $\int x^2 \sqrt{9 - 4x^2} \, dx$

Extra Example

Evaluate $\int \sqrt{2x^2 - 4} \, dx$